

Now that the first alien appears correctly, we'll write the code to draw an entire fleet.

Building the Alien Fleet

To draw a fleet, we need to figure out how to fill the upper portion of the screen with aliens, without overcrowding the game window. There are a number of ways to accomplish this goal. We'll approach it by adding aliens across the top of the screen, until there's no space left for a new alien. Then we'll repeat this process, as long as we have enough vertical space to add a new row.

Creating a Row of Aliens

Now we're ready to generate a full row of aliens. To make a full row, we'll first make a single alien so we have access to the alien's width. We'll place an alien on the left side of the screen and then keep adding aliens until we run out of space:

alien
_invasion.py

```
def _create_fleet(self):
    """Create the fleet of aliens."""
    # Create an alien and keep adding aliens until there's no room left.
    # Spacing between aliens is one alien width.
    alien = Alien(self)
    alien_width = alien.rect.width

    ❶ current_x = alien_width
    ❷ while current_x < (self.settings.screen_width - 2 * alien_width):
    ❸     new_alien = Alien(self)
    ❹     new_alien.x = current_x
        new_alien.rect.x = current_x
        self.aliens.add(new_alien)
    ❺     current_x += 2 * alien_width
```

We get the alien's width from the first alien we created, and then define a variable called `current_x` ❶. This refers to the horizontal position of the next alien we intend to place on the screen. We initially set this to one alien width, to offset the first alien in the fleet from the left edge of the screen.

Next, we begin the `while` loop ❷; we're going to keep adding aliens *while* there's enough room to place one. To determine whether there's room to place another alien, we'll compare `current_x` to some maximum value. A first attempt at defining this loop might look like this:

```
while current_x < self.settings.screen_width:
```

This seems like it might work, but it would place the final alien in the row at the far-right edge of the screen. So we add a little margin on the right side of the screen. As long as there's at least two alien widths' worth of space at the right edge of the screen, we enter the loop and add another alien to the fleet.